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Smart environmental DNA sampling strategies: a spatial autocorrelation method and interactive planner for survey design

Running headline: Efficient eDNA survey design

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Key words: biodiversity monitoring; decision-support tool; environmental DNA; fisheries management; Gaussian process; metabarcoding; sampling design; spatial autocorrelation; survey cost-effectiveness

Abstract

Environmental DNA (eDNA) surveys are increasingly used to monitor marine biodiversity, yet managers who must budget and justify surveys have little quantitative guidance on how many samples to collect, and because species differ in patchiness, no single sample size serves all. We hypothesized that a taxon's spatial autocorrelation length-scale (ρ), which describes how quickly its eDNA field changes across space, can guide this decision, because smoothly varying fields can be reconstructed from fewer samples than patchy ones. To test this, we simulated Gaussian process (GP) fields spanning a wide range of ρ , fitted Bayesian GP models to learn their parameters, progressively removed samples and refitted the models, and compared the resulting predictions and parameter estimates with those from the complete fields. Prediction error declined with both sampling density and ρ : at 6 samples per 10,000 km², log-scale error was ~ 1 (a factor of ~ 2.7) for $\rho = 50$ km but < 0.4 (~ 1.5) for $\rho > 500$ km, and covariance-parameter recovery stabilized above this density. This relationship is summarized in an invertible error–effort equation that converts a target prediction error and a prior estimate of ρ into a minimum sampling density and is implemented in a free, browser-based survey planner. We then applied the method in a case study combining eDNA data for 12 fish species (358 samples from five depths along the U.S. West Coast) with 200 simulated fields ($\rho = 25$ –1000 km). Species length-scales ranged from 22 to 1283 km. Error patterns in the simulations closely matched those in the observed data. Applied retrospectively, the planner showed that the 358 samples were sufficient to reach a target error of 0.5 for only the two smoothest species (nearly so for a third); the other nine needed 2.2–19.8 times as many. Together, these results show that the sampling effort needed to characterize a species' distribution is not universal but depends on its spatial autocorrelation, and can therefore be

specified in advance. The planner lets survey designers set defensible, species-specific effort before a cruise.

Introduction

Biodiversity monitoring programmes underpin decisions about where to fish, what to protect and how to report progress against national and international conservation targets, yet the budgets available to them are effectively fixed. How many samples to collect, and where, is therefore a management decision as much as a statistical one (Yoccoz et al., 2001). Survey design must balance the need for sufficient data for detecting trends and changes in population while also minimizing cost (i.e., maximizing sampling efficiency) (Stein & Ettema, 2003; Xu et al., 2015; Zhao et al., 2022).

The emergence of environmental DNA (eDNA) sampling has revolutionized monitoring capabilities, offering a non-invasive method to detect species presence and estimate relative abundance (Gold et al., 2021; Shelton et al., 2022; Taberlet et al., 2012). Recent synthesis work shows that eDNA methods have matured sufficiently for key uncertainties to be quantified and incorporated into existing fisheries stock-assessment frameworks and species distribution models, enabling the development of cost-effective abundance indices that complement conventional surveys (Baetscher et al., 2025; Guri et al., 2024; Kirtane et al., 2021; Salter et al., 2026). Because agencies are now evaluating eDNA as an operational component of biodiversity assessment and species distribution modeling rather than as an exploratory tool, explicit and defensible guidance on how much to sample has become a prerequisite for uptake rather than a refinement of it.

While increased sampling intensity generally reduces variance in population estimates and improves the precision of spatial pattern detection in species distribution models (Schweiger et al.,

2016), this relationship exhibits diminishing returns where beyond a certain threshold, additional sampling effort produces marginal gains in statistical precision that may not justify the associated costs and logistical demands (Perret et al., 2022; Zhao et al., 2022).

We conceptualize species spatial distributions as realizations of a Gaussian Process (GP), in which spatial dependence is characterized by a species-specific length-scale parameter (ρ) describing the distance over which observations remain correlated. A species with a long length-scale varies gradually across space and can therefore be reconstructed from fewer samples, whereas a species with a short length-scale exhibits more rapid spatial variation and consequently requires denser sampling (Figure 1). Thus, knowledge of species-specific spatial autocorrelation can provide a quantitative basis for informing sampling effort (Fortin et al., 1989). Yet, although general design guidance exists, for example on spatially balanced allocation of samples (Perret et al., 2022), on designs for estimating spatial autocorrelation (Laroche, 2024) and on eDNA sampling designs for fish diversity in lakes (García-Machado et al., 2024), there is still no quantitative, transferable procedure that returns how many eDNA samples a survey needs to reach a given precision, and design guidance is rarely used unless it is accessible to those who plan surveys. Here we used simulations to quantify how spatial autocorrelation and sampling effort affect the predictive accuracy of species distribution models and secondary measure the samples needed for estimating accurate of GP parameter. We then test whether the error structure of the simulations holds in real eDNA fields, using data for 12 fish species from the California Current, and use the same data to provide reference length-scales and to show how the planner can be used to judge whether an existing survey was large enough. Our objective is deliberately operational, namely to provide survey planners with guidance that can be applied before a cruise, based on a prior estimate of the spatial autocorrelation of target taxa and the level of precision required. We describe the

underlying method and the planner, evaluate them on simulated and empirical data, and outline applications to survey planning, monitoring and management.

Description of the method

The method links sampling effort to the accuracy of spatial inference within a common GP framework and proceeds in four steps, Step 0 to Step 3 (Figure 2): (0) generate simulated eDNA fields from a GP model across a wide range of spatial autocorrelation length-scales, so that the spatial structure of each field is known by construction and effort can be varied over a range no field campaign can achieve; (1) fit the same GP model to each complete simulated dataset to learn its governing parameters and to predict a reference field across a fixed prediction grid; (2) remove samples at progressively lower sampling densities and refit the GP model to each reduced-effort dataset and compare the parameters and the predicted fields with those obtained in Step 1, quantifying how parameter recovery and prediction accuracy degrade as effort is reduced; and (3) summarize the resulting prediction error as a function of sampling density and spatial autocorrelation in a regression that can be inverted to return the sampling density required for a user-defined error tolerance. The resulting equation is implemented in the “eDNA survey planner”, so that end users need only a prior estimate of spatial autocorrelation and a target error. The same workflow is also applied to observed eDNA data to benchmark the simulations against real fields. We describe each step generically, giving the values used in the case study where relevant. The results of the simulations are reported first (Method performance), followed by the empirical case study, which checks the simulations against real fields and demonstrates the planner on an existing survey.

Modelling framework: Gaussian processes

The GP approach is grounded in the assumption that species distributions and abundances change in a smooth and continuous manner across space (or other covariates), reflecting correlated responses to spatial, environmental, and biological covariates (Lewy & Kristensen, 2009). At the core, in simple terms, GPs posit that observations made at proximate locations or covariate values should be similar (Dormann et al., 2007). We assert that many species can be well described using relatively simple covariance models (Thorson et al., 2015). Specifically, we use an isotropic, squared exponential covariance model for the distribution of a species. Let $X = \{x_1, x_2, \dots, x_N\}$ be the set of N observed spatial locations. Then species (eDNA) abundance or density Y across all sampled locations (X) is modeled as,

$$Y \sim MVN(\mu, \mathbf{K} + \sigma^2 \mathbf{I}), \quad 1$$

where μ is the global mean abundance, σ^2 is the non-spatial variance, $\mathbf{I}$ is an $n \times n$ identity matrix and $\mathbf{K}(X, X)$ is the covariance matrix whose $(i, j)^{\text{th}}$ entry is given by the squared exponential kernel function,

$$\mathbf{K}(X, X) = k(x_i, x_j) = \alpha^2 \exp\left(-\frac{(x_i - x_j)^2}{2\rho^2}\right) \quad 2$$

where α^2 is the amplitude (i.e., the marginal variance of the spatial process, controlling the overall magnitude of variation in abundance), ρ is the length-scale parameter (in $x_i - x_j$ units; hereafter spatial autocorrelation).

Step 0: Generating the simulated spatial field

We generated in-silico eDNA concentration data using the GP model (Eq. 1-2) across the same spatial domain and prediction grid (Q) used in the empirical analysis. We defined this grid (Q) as 18,916 discrete point estimates, representing the centroids of approximately 8×8 km spatial cells

distributed across five depth layers. The number of points differed among depth layers (4,000, 4,000, 4,000, 3,622, and 3,294 points, respectively) because deeper layers covered a smaller spatial extent than shallower layers due to bathymetry.

The simulations were designed to explore a wide range of spatial structures by varying the GP length-scale parameter (ρ) across ten levels: 25, 50, 75, 100, 200, 300, 400, 500, 800, and 1000 km (Appendix S1: Table S1). For each value of ρ we generated (arbitrarily) 20 independent replicate spatial fields, resulting in a total of 200 simulated datasets (10 ρ levels $\times$ 20 replicates; Appendix S1: Table S1). Similar to GP inference of the empirical data, within each simulation, ρ was held constant across depth layers, but each layer had a separate mean (μ_d) drawn randomly from a normal distribution with mean of 0 and a standard deviation of 3. We focused only on spatial autocorrelation across latitude and longitude, allowing the eDNA concentration at each depth to be independent. The amplitude (α^2) and the non-spatial variance (σ^2) in Eq. 1 and 2 were fixed at 4 and 3, respectively, to mirror the empirical estimates.

Given the large number of projected eDNA points (Q), fitting a GP model using all locations ($\approx$ 20,000 input values) was computationally prohibitive. We therefore drew a subset of 4,000 points from each of the 200 simulated datasets as the “complete” observation dataset (Y) and its covariate (X), which constitutes the data used for all downstream analyses.

Step 1: Estimating GP-field spatial structure

Given simulated log eDNA concentration values (Y) paired with spatial coordinates (X) for each sampling location (x ; $N = 4,000$ per simulated field), we used the GP model (Eq. 1–2) to infer the

underlying spatial structure and learn key parameters governing the spatial field, including the depth-specific mean (μ_d), spatial autocorrelation (ρ), amplitude (α^2), and non-spatial variance (σ^2). To predict the eDNA concentrations (Z) across the spatial prediction grid (Q) for a given simulated field, we compute the covariance matrix among observed locations, $\mathbf{K}(X, X')$, using the squared exponential kernel and the non-spatial variance term ($\sigma^2 \mathbf{I}$). Using the Cholesky decomposition we solved for $(\mathbf{K}(X, X') + \sigma^2 \mathbf{I})^{-1}(Y - \mu_d)$ and we combined it with the cross-covariance between observed and prediction points, $\mathbf{K}(X, Q)$, to yield the predictions (Z) at each spatial point (Q):

$$Z = \mu_d + \mathbf{K}(X, Q)^\top (\mathbf{K}(X, X') + \sigma^2 \mathbf{I})^{-1} (Y - \mu_d) \quad 3$$

The resulting predicted field (Z) served as the “true” spatial field against which all subsequent reduced-effort predictions were compared (Figure 2A).

Step 2: Estimating prediction error under reduced sampling effort

To evaluate how sampling effort (E) influences spatial reconstruction and prediction accuracy, we generated a series of reduced-effort observation datasets (Y_E coupled with X_E) by randomly thinning the observation set (Y) and its spatial covariate (X) to fixed sample sizes (Figure 2). Each reduced-effort observation dataset (Y_E) together with its covariate (X_E) was refit with the same GP model (Eq 1-2) and produced effort-specific spatial predictions (S_E) on the full grid Q , by substituting S_E for Z , Y_E for Y and X_E for X in Eq. 3. This framework enables direct, like-for-like comparisons of reconstructed fields across sampling intensities since the predicted field is the same (Q). Sampling effort (E) was quantified as sampling density:

$$E = \frac{N}{A \times H} \times 10000 \quad 4$$

where A is the spatial area of the study domain ($\sim 200,000 \text{ km}^2$), and H is the number of depth layers (d ; $H = 5$), yielding E in units of samples per $10,000 \text{ km}^2$ per depth layer.

For each model prediction, we calculated the prediction error (ε), defined as the root mean squared error between spatial fields predicted from the full dataset (Z) and those predicted from each reduced sampling scenario (S_E) as follows (Figure 2):

$$\varepsilon_{\text{sim}} | E = \sqrt{\frac{1}{n} \sum_{i=1}^n (Z_i - S_{iE})^2} \quad 5$$

with n being the total number of Q grid points ($n = 18,916$) where the predictions (Z and S_E) are projected. Here, $\varepsilon_{\text{sim}} | E$ denotes the prediction error of the simulated fields at sampling density E , and $\varepsilon_{\text{obs}} | E$ the prediction error of the observed data at sampling density E ; hereafter we write ε_{sim} and ε_{obs} .

Additionally, to evaluate the influence of sampling effort on GP parameter estimation, we compared the estimated parameters ($\rho, \mu_d, \sigma^2, \alpha^2$) from effort-specific datasets (Y_E) and covariates (X_E) to the true values used to generate the simulated fields. Differences between estimated and simulated parameter values were quantified using the root mean squared error (hereafter τ).

Step 3: Linking prediction error to sampling effort and spatial autocorrelation

We modeled prediction error (ε) as a function of sampling density (E) and spatial autocorrelation (ρ), including their interaction using a linear model as follows:

$$\log(\varepsilon) = \omega + \beta \log(E) + \theta \log(\rho) + \gamma \log(E)\log(\rho) \quad 6$$

where ω is the intercept, β slope of sampling density (E), θ is the slope of spatial autocorrelation (ρ) and γ is the coefficient on the interaction between the two latter factors.

Solving for the sampling density

For a given ρ and a stated tolerance for prediction error (ϵ^* , the level of error the manager is willing to accept), Eq. 6 reduces to a power law in E that can be inverted to return the minimum sampling density (E^*) required to achieve that target error:

$$E^* = \exp\left(\frac{\log(\epsilon^*) - \omega - \theta \log(\rho)}{\beta + \gamma \log(\rho)}\right) \quad 7$$

Eq. 7 is deterministic: it returns a single sampling density for a given ρ and ϵ^* , and it describes the expected relationship between effort, spatial autocorrelation and error rather than the scatter around it. Realized errors in any individual survey will therefore vary around ϵ^* , and the range of sampling densities compatible with a given target is wider than the single value returned. The same relationship can also be run in reverse, so that a fixed budget of N_H samples per depth supports a survey area of $10,000 N_H/E^*$ at the target error. Eq. 7 is calibrated over $\rho = 25\text{--}1000$ km and $E = 1.5\text{--}20$ samples per $10,000 \text{ km}^2$ per depth layer, and results outside this range are extrapolations. All the GP models (equation 1 and 2) were fitted in a Bayesian framework using Stan probabilistic programming language (Stan Development Team, 2023)10/8/2026 6:12:00 PM, with weakly informative priors, and the model in Eq. 6 was fitted using the *lm()* function in R (R Core Team, 2024); sampler settings, convergence diagnostics and prior distributions are given in Appendix S1: Section S3 and Table S2.

Method performance: simulation results

We evaluated the method on the 200 simulated fields generated in Step 0. Here we report how the prediction error and the recovery of the Gaussian process parameters change with sampling density and spatial autocorrelation, and how the prediction error is summarized in the error–effort model (Eq. 6) that underlies the planner.

Prediction error and the error–effort model

Prediction error in simulation data (ε_{sim}) decreased non-linearly with increasing sampling effort and with higher spatial autocorrelation (Figure 3A). These patterns were supported by a log–log linear model where the prediction error (ε) decreased exponentially for every exponential increase in sampling density ($\beta = -0.176$) and spatial autocorrelation ($\theta = -0.228$), with an additional negative interaction ($\gamma = -0.0684$) and with an intercept of $\omega = 1.65$. For instance, at a sampling intensity of 6 samples per 10,000 km², species with spatial autocorrelation values of 50 km had mean prediction errors of 1 (Figure 3A) indicating that on average every point predicted the eDNA concentration will differ from the predicted true concentration (Z) by a factor of 2.72 (for perspective a concentration of 100 copies/L will be predicted between 37-270 copies/L; Appendix S1: Figure S3). In contrast, for the same sampling effort (i.e., 6 samples per 10,000 km²) but species with spatial autocorrelation $\rho > 500$ km had errors below 0.4 indicating an average of over- and underestimation by a factor of 1.5 (for perspective a concentration of 100 copies/L will be predicted between 67-150 copies/L; Appendix S1: Figure S3). Importantly, diagnostic checks revealed no systematic bias in the predictions, so the observed variation reflects pure precision (Appendix S1: Figure S4).

GP parameter recovery

Sampling effort and spatial autocorrelation strongly influenced the ability of the GP model to recover the parameters used to generate the simulated spatial fields (Figure 4). Estimation error for the depth-specific mean (μ_d) decreased with increasing sampling density for simulated fields with small spatial autocorrelation but showed the opposite pattern for fields with large spatial

ranges (Figure 4C). Specifically, τ (root mean squared error between estimated and simulated) for μ_d increased with sampling effort when spatial autocorrelation exceeded 400 km, indicating that mean estimates became increasingly biased in highly spatially structured fields. Additionally, the recovery of the spatial autocorrelation length-scale (ρ) was highly sensitive to sampling density (Figure 4A). At low sampling effort ($E < 4$), estimates of ρ showed substantial error, particularly for simulations with large spatial ranges. However, estimation accuracy improved rapidly as sampling effort increased, with τ declining sharply and stabilizing once sampling densities were higher than 6 samples per 10,000 km², beyond which additional sampling produced only marginal improvements in ρ estimation.

Estimation error for the amplitude parameter (α^2) declined with increasing sampling density for simulations with small spatial autocorrelation, while remaining relatively stable across sampling densities for simulations with larger spatial ranges (Figure 4D). Overall improvements in α^2 estimation were therefore limited once sampling effort increased above 6 samples per 10,000 km². The non-spatial variance parameter (σ^2) was consistently well recovered across sampling densities (Figure 4B). τ declined sharply with increasing sampling density and approached near-zero values once sampling densities exceeded 6 samples per 10,000 km².

Implementation: the eDNA Survey Planner

To make Eq. 6–7 usable without a prior installation of a statistical software, we implemented them in the “eDNA survey planner”, a browser-based application that can run offline (Appendix S1: Figure S6). All calculations run locally in the browser, so no survey data are uploaded, and the only online component (a basemap for drawing the study domain) is optional, as boundaries can also be entered as coordinates; an R version with the same functions is also provided with the

272 archived code. The planner answers the two questions survey designers most often face. “How
 273 many samples?” takes a target error (ε^*), a prior estimate of ρ (entered directly or chosen from
 274 presets ranging from very patchy, 25 km, to very smooth, 1,000 km), the survey area per depth
 275 (measured from a study domain of 1,000–1,000,000 km² per depth layer that the user outlines on
 276 a map, loads from the Pacific example used in the case study or enters as coordinates) and the
 277 number of depth layers, and returns the required sampling density, the samples per depth and the
 278 total site–depth observations. Additionally, an optional “How much area?” calculation takes ε^* , ρ
 279 and a sample budget per depth, and returns the area that budget can support at the target error.
 280 Because ε is a root mean squared error on the natural-log scale, the planner translates it into a
 281 concentration range: around a reference concentration C , predictions typically fall within
 282 $\exp(\log(C) \pm \varepsilon)$, a multiplicative factor of e^ε . An interactive error–effort curve shows where the
 283 target lies on the curve of diminishing returns, and a two-dimensional GP simulation displays, for
 284 the chosen domain, a reference field, the samples required to reach ε^* and the resulting
 285 reconstruction, together with a map of its pointwise prediction error. This simulation is a visual
 286 demonstration of the recommended survey and does not enter the calculation of the sampling
 287 density.
 288 Default coefficients are those of Eq. 6 estimated from the simulations, but they can be edited, so
 289 that programs can substitute a calibration for their own region, taxa or assay by applying Steps 0–
 290 3 with the archived code. Results outside the calibrated range are calculated but flagged as
 291 extrapolations, and the planner states that ε is an average error across the survey domain rather
 292 than a percent error, a detection probability or a pointwise confidence interval. Recommendations
 293 follow the central fitted curve; uncertainty in the coefficients is not yet propagated. The output is
 294 therefore an exploratory planning aid rather than a guarantee that a given ε will be achieved in a

real survey, and the area returned from a sample budget is a density-equivalent value rather than a boundary of reliable inference.

Case study: a multispecies eDNA survey of the California Current

Study system and eDNA data

Environmental DNA samples were collected during the Joint U.S.-Canada Integrated Ecosystem and Pacific Hake Acoustic-Trawl Survey aboard the NOAA ship *Bell M. Shimada* in summer 2019, from central California to northern Washington (38.36°–48.46° N, 123.53°–125.93° W; study domain of approximately 200,000 km²). Seawater was collected with Niskin bottles at five target depths (0, 50, 150, 300 and 500 m) and filtered onboard, yielding 358 site–depth samples from 152 sampling locations. Species-specific eDNA concentrations (log copies/L) were estimated for 12 fish species by jointly modeling 12S rRNA metabarcoding reads (MiFish primers), a mock community to correct for amplification bias and qPCR measurements of Pacific hake (Guri et al., 2024, 2025). The focal assemblage comprised pelagic, mesopelagic and demersal taxa characteristic of the California Current ecosystem: *Clupea pallasii*, *Engraulis mordax*, *Leuroglossus stilbius*, *Microstomus pacificus*, *Sardinops sagax*, *Scomber japonicus*, *Sebastes entomelas*, *Stenobranchius leucopsarus*, *Tactostoma macropus*, *Tarletonbeania crenularis*, *Thaleichthys pacificus* and *Trachurus symmetricus*. Sampling, laboratory and bioinformatic details are given in Appendix S1: Section S1.

Spatial structure of the studied species (Step 1)

As in Step 1 of the simulations, we fitted the GP model (Eq. 1–2) to the log eDNA concentrations of each of the 12 species ($N = 358$ site–depth samples) and used Eq. 3 to predict eDNA concentrations across the prediction grid (Q) defined in Step 0.

Across species, depth-specific global mean eDNA concentrations generally declined with increasing sampling depth (Figure 5). Species-specific length-scale estimates varied considerably across species (Figure 5A), likely reflecting differences in habitat use and movement behavior. Surface-dwelling species tended toward shorter spatial autocorrelation lengths and showed more pronounced differences between surface and deep mean eDNA concentrations (μ ; Figure 5B), though exceptions were evident. Deep-dwelling species broadly exhibited longer spatial autocorrelations (Figure 5A) and relatively stable mean eDNA concentrations across depths, although this pattern was not universal. *Tactostoma macropus*, for instance, showed a comparatively short length-scale despite its deep-water affinity. Estimated length-scales ranged from 22 to 1283 km across the 12 species (Table 1).

Estimating prediction error under reduced sampling effort (Step 2)

As in Step 2, we randomly thinned the 358 observations to $N = 330, 300, 260, 220$ and 150 (Figure 2B), refit the GP model to each reduced dataset and calculated the prediction error (ϵ_{obs}) against the field predicted from the full dataset as in Eq. 5, repeating the procedure across 20 random realizations for each thinning intensity.

Prediction error in empirical data decreased with increasing sampling density across all species (Appendix S1: Figure S1 and S2). However, this reduction was strongly modulated by the degree of spatial autocorrelation, similar to the simulations (Appendix S1: Figure S2). Species

characterized by short spatial autocorrelation (small ρ values) consistently exhibited higher prediction error (ε_{obs}) across all sampling densities, whereas species with wider spatial autocorrelation structures (e.g., $\rho > 500$ km) showed substantially lower prediction errors (ε_{obs}) even under comparatively sparse sampling regimes.

Linking prediction error to sampling effort and spatial autocorrelation (Step 3)

As in Step 3 of the simulations, we fitted Eq. 6 to the observed prediction errors (ε_{obs}), using the length-scale estimated for each species in Step 1 (ρ) and the sampling density of each thinned dataset (E), to describe how prediction error in real eDNA fields depends jointly on sampling effort and spatial autocorrelation.

The observed and simulated prediction errors cannot be compared directly because they are measured against reference fields of different quality: ε_{obs} is measured against the field predicted from the 358 observed samples, whereas ε_{sim} is measured against the field predicted from 4,000 samples (Figure 2). To put the simulations on the same footing as the observed data, we drew 358 samples from each simulated field, used the field predicted from these 358 samples as the reference, thinned them to the same sample sizes as the observed data ($N = 330, 300, 260, 220$ and 150) and calculated the resulting prediction error ($\varepsilon_{\text{sim},358}$) as in Eq. 5. We then fitted Eq. 6 to $\varepsilon_{\text{sim},358}$ and compared its coefficients with those fitted to ε_{obs} , as close agreement would indicate that the simulations reproduce the error structure of real eDNA fields (Appendix S1: Section S2).

The fitted coefficients showed the same directional effects as in the simulations, with prediction error decreasing with increasing sampling density (β) and with increasing spatial autocorrelation (θ), and a negative interaction between the two (γ). The estimates of all four coefficients, the intercept (ω) and the three slopes (β , θ and γ), also aligned closely with the 1:1 line, demonstrating

a high degree of structural agreement between the empirical and simulated error models (Appendix S1: Figure S1).

Applying the planner

To show how the planner is used, we asked whether the 358 site–depth samples collected in 2019 ($E = 3.6$ samples per 10,000 km² per depth layer) were enough to reach a target prediction error of $\varepsilon^* = 0.5$ (an arbitrary value chosen for illustration), corresponding to predictions within a typical factor of about 1.65 of the reference field. For each species we entered its estimated length-scale (Step 1 applied to the observed data), the survey area ($\sim 200,000$ km² per layer) and the five depth layers into Eq. 7, which returns the required sampling density (E^*) and the corresponding total number of samples (Table 1). The results showed that the samples collected were sufficient for the two species with the longest length-scales, *Leuroglossus stilbius* and *Microstomus pacificus*, for which the planner required 291–295 samples, about 17–19% fewer than were collected, and were close to the requirement for *Sebastes entomelas* (373 samples; $\rho = 763$ km). For the remaining 9 species the planner required 2.2 to 19.8 times as many samples as were collected, and the requirement rose steeply as the length-scale decreased (for example 1,011 samples for *Clupea pallasii*, $\rho = 150$ km, and 7,078 for *Tarletonbeania crenularis*, $\rho = 22$ km). Because five species have length-scales outside the calibrated range ($\rho = 25$ –1000 km) and four require more than the calibrated maximum of 20 samples per 10,000 km² per depth layer (Table 1), the largest requirements are extrapolations. They indicate that the 2019 effort fell far short of the target for these species, but they should not be read as precise sample sizes.

384 **Table 1.** Reference spatial autocorrelation length-scales (ρ) for 12 fish species in the California
385 Current, and the sampling required to reach a target prediction error of $\varepsilon^* = 0.5$ (Eq. 7; 200,000
386 km² per layer, five depth layers). E^* is the required sampling density (samples per 10,000 km² per
387 depth layer), N^* the total number of site–depth samples required ($E^* \times 100$), and the last column
388 compares N^* with the 358 samples collected (values below 1 indicate that the collected samples
389 exceeded the requirement). † Outside the calibrated range of the error–effort model ($\rho = 25$ –1000
390 km or $E = 1.5$ –20 samples per 10,000 km² per layer): extrapolation.

Species	ρ (km)	E^*	N^*	$N^* / 358$
Leuroglossus stilbicus†	1283	2.9	291	0.81
Microstomus pacificus†	1243	3.0	295	0.83
Sebastes entomelas	763	3.7	373	1.04
Engraulis mordax	207	8.0	804	2.25
Sardinops sagax	181	8.8	882	2.46
Clupea pallasii	150	10.1	1,011	2.82
Scomber japonicus	89	15.4	1,540	4.30
Tactostoma macropus	79	17.0	1,702	4.75
Stenobranchius leucopsarus†	41	32.8	3,283	9.17
Trachurus symmetricus†	24	60.8	6,076	17.0
Thaleichthys pacificus†	23	65.3	6,532	18.2
Tarletonbeania crenularis†	22	70.8	7,078	19.8

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Discussion

Translating species-specific spatial autocorrelation into sampling effort provides the quantitative, transferable basis for eDNA survey design that has so far been lacking. Prediction error declined with both sampling density and spatial autocorrelation (Figure 3A), whereas the recovery of GP parameters was much less sensitive to effort (Figure 4), and the same error structure held in real eDNA fields (Appendix S1: Figure S1). Optimal sampling effort is therefore not universal but depends on the spatial autocorrelation of the target species and on the precision a survey requires.

Sampling effort, spatial autocorrelation and prediction accuracy

The value of additional sampling effort depends on the spatial structure of the target species (Xu et al., 2015; Zhao et al., 2022). Homogeneously distributed species are inherently easier to reconstruct than patchier ones at equivalent effort (Wagner & Fortin, 2005). Additionally, acceptable error is context dependent. Studies requiring high-resolution quantitative estimates may justify stricter precision, whereas broad-scale biodiversity screenings may tolerate looser precision and sparser sampling (Perkins et al., 2021; Radinger et al., 2019). Once the required precision and ρ are known (from prior literature, pilot sampling or expert judgement), the fitted log–log model (Eq. 6; Figure 3A; Appendix S1: Figure S5) can be inverted (Eq. 7) to solve for the minimum sampling density E required to reach a target error, a calculation the planner performs interactively.

Although effort is reported per 10,000 km², due to its dependence on the study domain and the number of samples, what matters most is how far apart samples are relative to the autocorrelation of a species. A sampling effort E places samples roughly $\sqrt[3]{(10,000/E)}$ km apart, so the effort needed for a given target error can equivalently be expressed as the ratio of sample spacing to

length-scale (d/ρ). In the 2019 survey ($E = 3.6$), spacing was about 53 km, giving a d/ρ ratio of roughly 2.4 for the patchiest taxon ($\rho = 22$ km) and 0.04 for the smoothest ($\rho = 1283$ km; Table 1). In the first case neighboring samples are separated by more than two length-scales and share little information, whereas in the second many samples fall within a single length-scale and carry overlapping information, which is why the same effort can be too sparse for one taxon and sufficient for another. Expressing effort relative to length-scale also makes the result easier to carry to domains of different size or to surveys with different numbers of depths, although the ratio is not a single fixed threshold because error depends on E and ρ jointly (Eq. 6), and its transferability beyond the domain and length-scales studied here remains to be tested.

Case study: agreement with observed data and implications

In the case study, the error model estimated from the observed data reproduced the structure of the model estimated from the simulations. This supports the use of the simulation-derived coefficients for taxa and domains similar to those studied here.

The case study is also the main limit on how far these conclusions can be generalized. It covers 12 fish species from a single region, season (summer 2019), survey and assay, within one domain of about 200,000 km², and the length-scales reported here describe the eDNA fields of that system, which are shaped by local hydrography as well as by the biology of each species. Agreement between the simulated and observed error models therefore shows that the approach is sound for this assemblage and for domains and taxa like it, not that the coefficients transfer unchanged to other taxa such as invertebrates or marine mammals, to other ecosystems such as estuaries, lakes or the open ocean, or to other seasons and sampling methods. The framework is built to test this directly, because the same workflow can be applied to any dataset using the archived code.

438

439 **Assumptions and limitations of the method**

440 A first limitation concerns how well ρ itself can be estimated at the sampling effort available in
441 the case study. The 358 samples correspond to $E = 3.6$ samples per 10,000 km² per depth layer,
442 below the density at which the recovery of ρ stabilized in the simulations (Figure 4A). At this
443 effort, length-scales may be over- or under-estimated at both ends of the range. For very patchy
444 taxa (ρ below roughly 50), spacing is comparable to or greater than the distance over which
445 samples remain correlated, so few sample pairs resolve how quickly correlation decays. For very
446 smooth taxa (ρ above roughly 500 km, including the two taxa whose estimates exceed the 1000
447 km upper limit of the simulations), the length-scale approaches the extent of the domain and
448 becomes difficult to separate from the depth-specific mean (Figure 4A, C). The shortest and
449 longest length-scales in Table 1 should therefore be read as indicative rather than precise. Because
450 ρ is the main input to the planner, errors in its estimate carry through to the effort the planner
451 returns, and users working with taxa at either extreme may wish to explore a range of plausible
452 length-scales rather than a single value.

453 Extrapolating directly from simulations to natural systems also requires caution because real
454 eDNA fields may violate the simplifying assumptions of the stationary, isotropic squared
455 exponential kernel. Spatial structure may vary across the domain (non-stationarity), differ by
456 direction (anisotropy), or include sharp transitions (fronts, plume boundaries, shelf-break
457 structure) that a globally smooth covariance function represents poorly (Paciorek & Schervish,
458 2006).

459 In addition, ecological processes such as migration, schooling, habitat discontinuities or patchy
460 prey fields can also generate multi-scale patterns that depart from a single characteristic length-

scale (Wagner & Fortin, 2005), and may help explain the elevated prediction error at intermediate autocorrelation levels in the empirical data, where such complexity is not represented in the simulations.

Lastly, observation error in eDNA data is unlikely to be purely Gaussian, arising from PCR bias, compositional constraints, incomplete calibration and heterogeneity in filtration or extraction efficiency (Bessey et al., 2020; Gloor et al., 2017; Kelly et al., 2019; Shaffer et al., 2025), but the modelling framework used here accounts for these sources (Guri et al., 2025), so they are unlikely to bias the results presented here.

Furthermore, the simulation assumes a stationary GP with fixed amplitude and noise and a single length-scale across depths, whereas real eDNA fields may have variable variance structure and dependence on environmental covariates such as temperature, salinity, chlorophyll and bathymetry (Liu et al., 2026). Our prediction error (ϵ_{sim}) is also computed against a "complete" prediction from a 4,000-point subset (Z) rather than the full grid, a limitation most visible in the secondary thinning comparison, where the reference field is constrained to $N = 358$ samples, making prediction error higher and more stochastic. In addition, Eq. 7 reflects the GP model used here, and models that survey teams build in the future from their own data (for example with different kernels, covariates, priors or observation models) can add further variance, so realized errors may exceed the target error (ϵ^*) even when the recommended sampling density is achieved.

Intrinsically, eDNA integrates ecological information through space and time via transport and decay (Brandão-Dias et al., 2023; Fukaya et al., 2021), so even where organisms are locally patchy the observed eDNA field may be smoother than the organism field, depending on hydrodynamic conditions and degradation rates (Brandão-Dias et al., 2025). Such integration is embedded within the inferred length-scale, so ρ should be interpreted as an effective length-scale of the observed

eDNA field rather than a purely ecological niche parameter (Jo et al., 2025). Integrating eDNA observations with independent abundance data (trawl, acoustics) would help separate these contributions, enabling clearer ecological interpretation of ρ and improving the transferability of sampling-effort recommendations (Guri et al., 2024). In a small domain, the autocorrelation scale of the observed eDNA field may exceed the survey extent, making the field appear nearly uniform and the depth-specific mean (μ_d) difficult to estimate independently of the spatial process, as observed in our simulations (Figure 4C) and reported previously (Paciorek, 2010; Redfern et al., 2006).

Perspectives and applications

The eDNA Survey Planner turns a single estimable quantity, the spatial autocorrelation of the target taxon, into a transparent calculation that can be run before a vessel leaves port. It needs only an outline of the study domain, a prior estimate of ρ and a target error, runs in a web browser (also offline) without installation and uploads no survey data, so it is accessible to the people who allocate ship time and not only to those who fit geostatistical models. It can be used in either direction, converting a target error into the sampling density a survey needs, or converting a fixed sample budget into the area that can be covered at that error. Because the same relationship can be applied to an existing design, the planner also supports retrospective use, as in the case study, where it showed how far a completed survey met a stated precision for each taxon.

For data collection, the planner lets survey teams state their assumptions about spatial structure and precision at the proposal stage and compare designs for taxa with different length-scales. The value of the planner depends on what a survey is meant to deliver. For stock assessment the key output is usually an aggregate abundance or biomass index, to which spatial structure contributes

mainly through the precision of that index (Thorson et al., 2015), and, an eDNA-derived abundance index was incorporated for the first time in the 2025 U.S.-Canada Pacific hake assessment (Johnson et al., 2025). The planner's error metric, which measures how well a whole field is reconstructed, is not the same as the uncertainty of an area-integrated total, so it should not be read as a direct measure of assessment precision. It applies directly where the spatial field itself is the product, as in distribution maps, range shifts, protected areas, bycatch avoidance and multispecies management (Pickens et al., 2021). In these cases sampling density is often the bottleneck, because conventional surveys are costly and cover space sparsely (Rufener et al., 2021; Vilas et al., 2025), whereas eDNA tends to lower the cost and effort of sampling (Iacaruso et al., 2025) and can therefore add many more sampling points. It is best seen as complementing conventional gear rather than replacing it (Guri et al., 2024; Hansen et al., 2018; Lacoursière-Roussel et al., 2016; Ramírez-Amaro et al., 2022; Salter et al., 2026).

For monitoring and management, the planner can be used to evaluate whether existing programs are matched to the objectives they are meant to serve, and to anticipate how effort may need to change if new target taxa are added. Such programs are required for marine protected areas and other regulated places, where increasing sampling effort has been shown to shorten the time needed to detect change (Perkins et al., 2021). Because the planner converts a target error and length-scale into a sampling density, it can be used to size baseline surveys and before-after-control designs. For programs under budget pressure or environmental change, the planner can evaluate adaptations before they are made. Survey time series can be disrupted when areas are closed for conservation or energy development, when stocks move beyond historical survey areas, or when funding, vessels or weather constrain data collection (Guri et al., 2024), and managers increasingly must decide where effort can be reduced without degrading data quality (DeFilippo et al., 2023).

Coupling molecular tools with conventional surveys can add flexibility (Guri et al., 2024), and the planner can specify how much eDNA sampling maintains a given precision where conventional gear can no longer be deployed. Because it returns a number of samples, it can also be combined with per-sample costs to produce a budget; analyzing ten eDNA samples cost roughly €5,000 in one global survey (Sanchez et al., 2025), and the planner complements frameworks that weigh survey costs against detection or accuracy (Lugg et al., 2018; Muradian et al., 2019; Smart et al., 2016).

For policy implementation, the planner offers a transparent and reproducible way to justify the sampling effort written into monitoring plans, survey proposals and funding requests, and its most direct use is in decisions that depend on knowing where species occur. Range maps are cornerstones of conservation strategies such as the classification of threatened species, yet occurrence records remain incomplete: global eDNA surveys extended the known geographic ranges of most marine fishes examined, with the chance of doing so depending mainly on prior sampling effort in an area (Sanchez et al., 2025). Spatially explicit eDNA maps can also identify biodiversity hotspots that coincide with documented regions of ecological significance, such as the Columbia River plume and Heceta Bank (Guri et al., 2025), and have been used to map the marine distribution of the eulachon a species listed as threatened under the U.S. Endangered Species Act, to augment recovery planning and potentially inform bycatch mitigation (Liu et al., 2026). Decisions about where to establish or adjust protected areas, closures or bycatch measures depend on how reliably such maps resolve the species of concern, and the planner makes that reliability explicit. Because patchy species need denser sampling than smoothly distributed ones to reach the same map accuracy, as the case study showed, a survey sized for the latter may be too sparse to delineate habitat for the former. Global targets for protected-area coverage emphasize the quantity

of area protected over whether biodiversity is represented within it, which can lead to inefficient conservation action (He et al., 2023). Stating the length-scale, target error and sampling effort behind a distribution map lets agencies and reviewers judge whether the map is robust enough to support a designation, and revisit boundaries as new data accumulate.

The planner is also designed to be adapted rather than applied as a fixed rule. Its coefficients can be edited, so that programs can substitute a calibration for their own region, taxa or assay by applying the workflow to their own data with the archived code, and the length-scales estimated here for 12 fish species can serve as starting priors until regional calibrations exist. Reporting estimated length-scales alongside concentration results would allow these priors to accumulate across surveys and regions. Natural extensions of the framework are to propagate uncertainty in the model coefficients into the recommended sampling density, to relax the stationary, isotropic kernel so that non-stationary, anisotropic or covariate-dependent spatial structure can be represented, and to link length-scales to independent abundance data so that they can be interpreted ecologically as well as statistically. Because the code and the planner are openly available, such extensions and regional calibrations can be developed and shared by the wider eDNA community.

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576

577 **Author contributions**

578 Gledis Guri conceived the study, developed the methodology and the survey planner, performed
579 the analyses and wrote the manuscript; Andrew O. Shelton contributed to the methodology,
580 funding acquisition, data interpretation and discussion; Ryan P. Kelly contributed to
581 conceptualization, funding acquisition, data interpretation and discussion; Owen R. Liu
582 contributed to data interpretation and discussion; Krista M. Nichols contributed to
583 conceptualization, funding acquisition and discussion. All authors contributed critically to the
584 drafts and gave final approval for publication.

585

586 **Conflict of interest statement**

587 The authors declare no conflicts of interest.

588

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Figure captions

Figure 1. Conceptual illustration of how sampling density affects the reconstruction of (A) a patchy and (B) a smooth concentration field along a one-dimensional transect (km). Top rows show the true concentration (black line); middle and bottom rows show reconstructions (red dashed line) from few and many samples (green circles), respectively.

Figure 2. Workflow of the method for simulated (A, *in silico*, true spatial field known) and real-world (B, *in situ*, true spatial field unknown) eDNA data. In Step 0 a spatial field with known parameters is generated on a discrete grid (Q , simulations only), in Step 1 the field is sampled (Y) and a GP model is fitted to predict the reference field (Z), in Step 2 the samples are thinned to fewer samples (Y_E) and the GP model is refitted to predict the field again (S_E), and in Step 3 the resulting prediction errors are related to sampling effort and spatial autocorrelation. Three prediction errors are calculated, ε_{sim} and $\varepsilon_{\text{sim},358}$ for the simulated data (using Z and the prediction from 358 samples as the reference, respectively) and ε_{obs} for the observed data, and each is fitted to a relationship with ρ and E (Eq. 6).

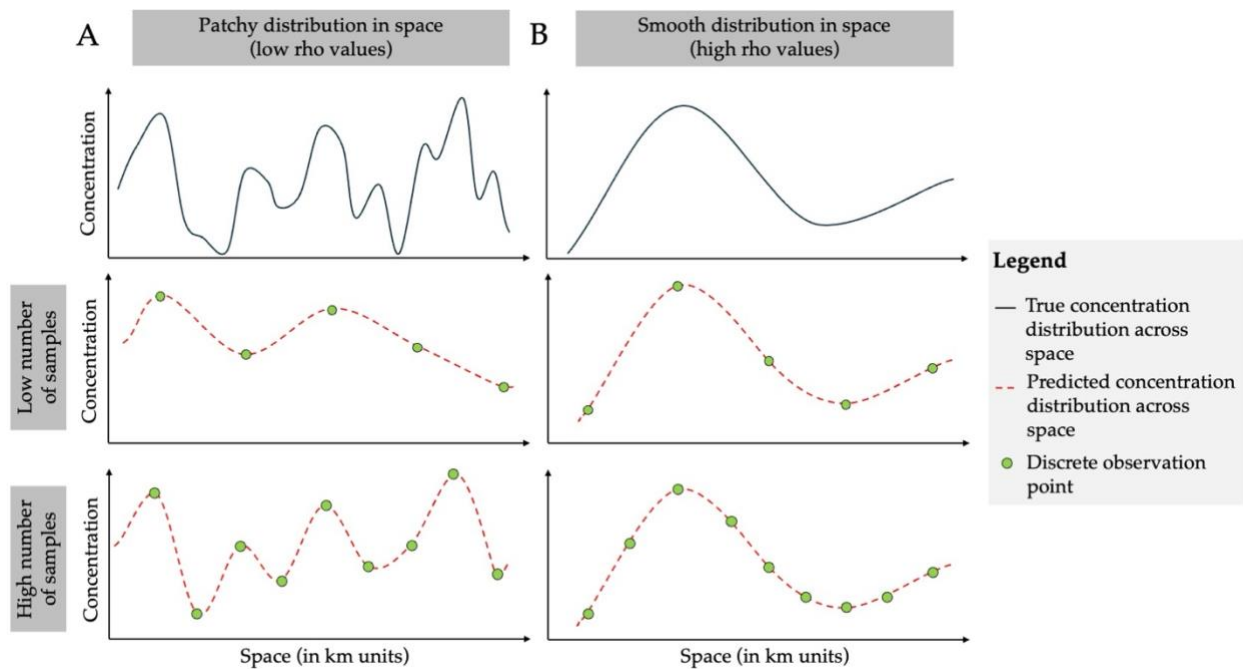
Figure 3. Prediction error (ε_{sim}) across sampling densities (E , samples per 10,000 km² per depth layer) for simulated fields with different length-scales (colors); points are simulation outcomes and lines mean trends.

Figure 4. Recovery of Gaussian process parameters in the simulations: mean root mean squared error (τ) between estimated and true values across 20 replicates for (A) ρ , (B) σ^2 , (C) depth-specific mean (μ_d) and (D) α^2 , by sampling density (E) and length-scale (colors).

Figure 5. Gaussian process parameter estimates for the 12 species. (A) Spatial autocorrelation length-scale (ρ , km) by depth association (surface, midwater, deepwater). (B) Depth-specific mean

808 eDNA concentration (μ ; log copies/L). (C) Amplitude (α^2) versus non-spatial variance (σ^2). Points
 809 are posterior means and bars 95% credible intervals; colors identify species.

810 Figure 1



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812 Figure 2

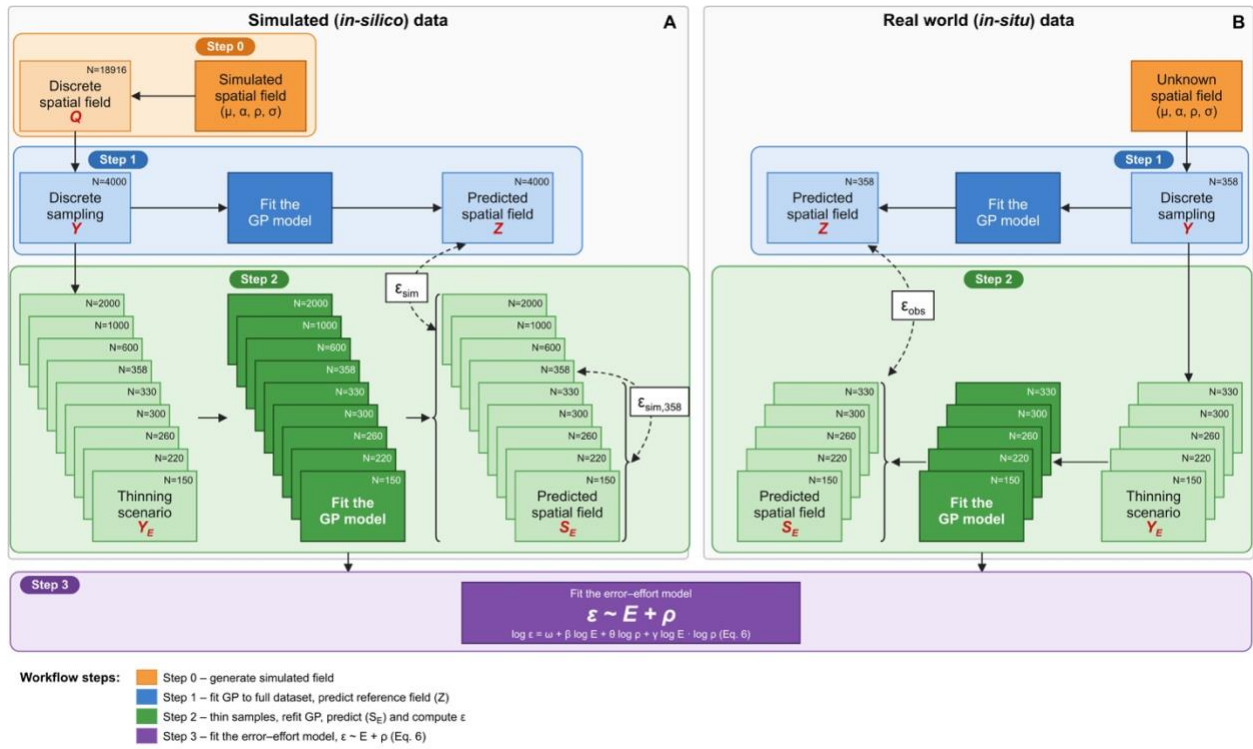


Figure 3

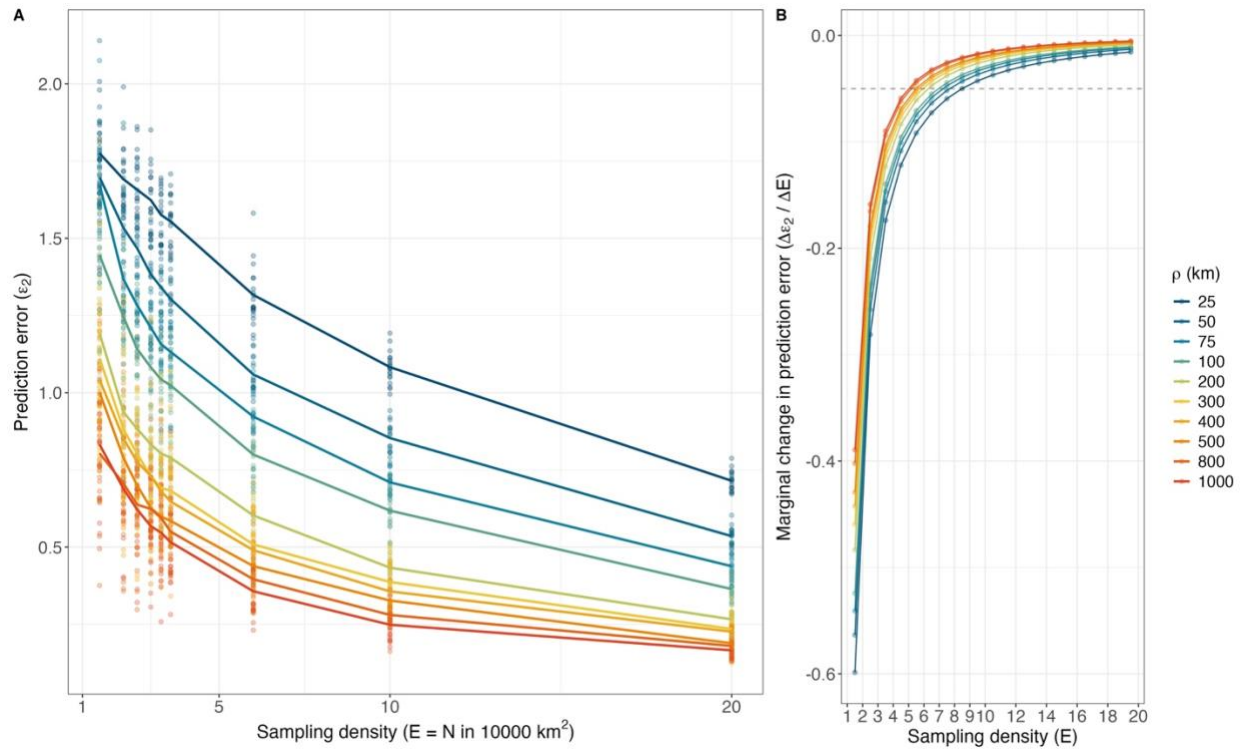
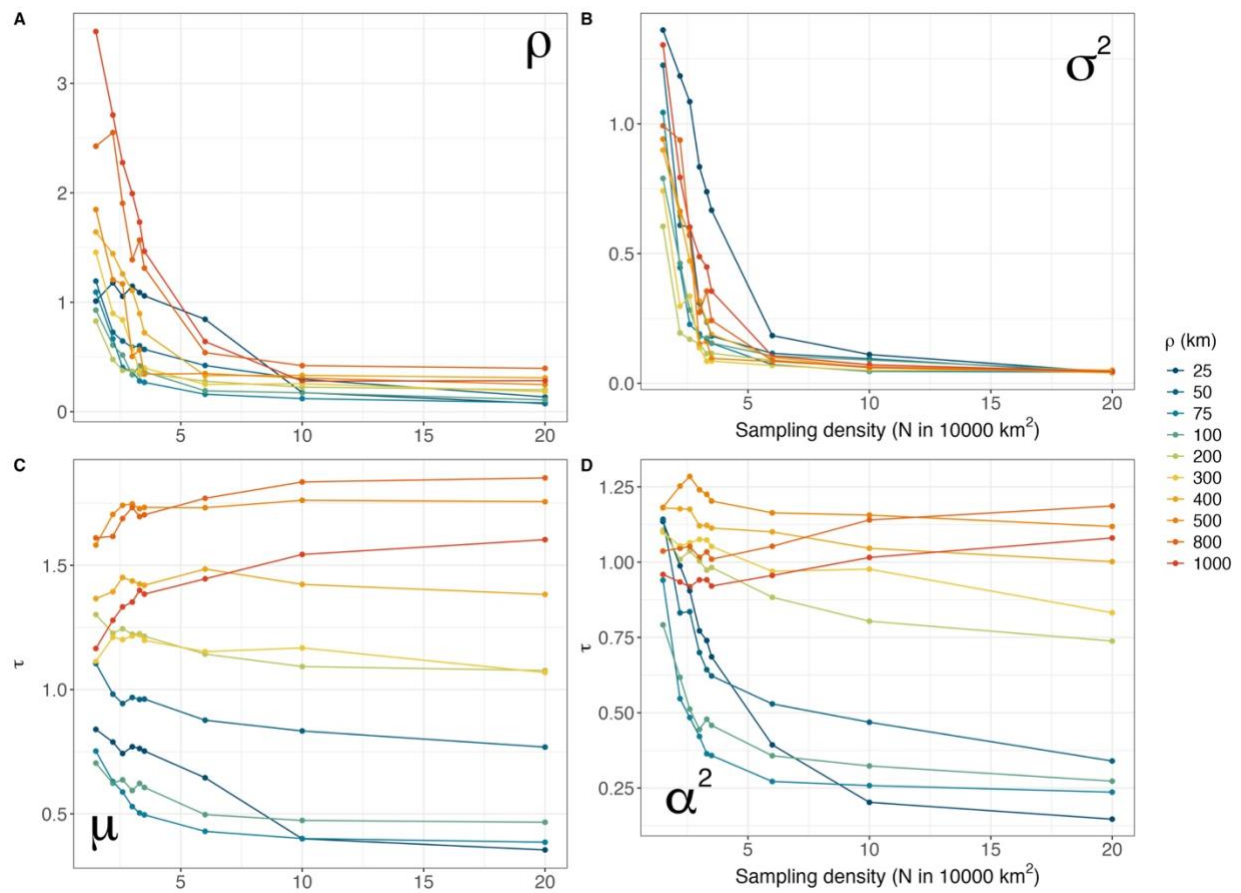
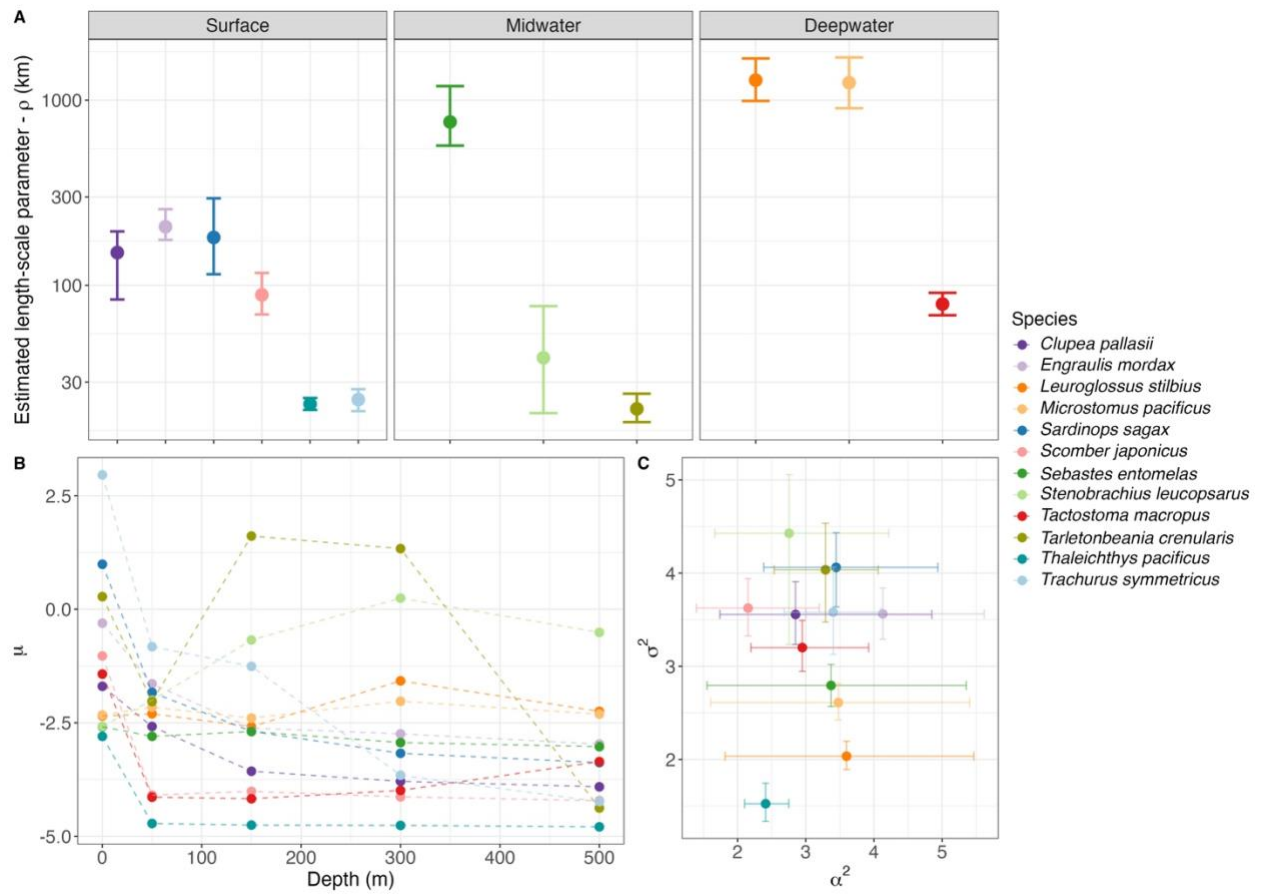


Figure 4



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818 Figure 5



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